

SONiC SmartFabric Manager for PowerStore

Deployment Guide

Abstract

This deployment guide covers using Dell Technologies SONiC SmartFabric Manager to configure a leaf-spine topology for single or multirack PowerStore NVMe/TCP storage deployments.

Notes, cautions, and warnings

 **NOTE:** A NOTE indicates important information that helps you make better use of your product.

 **CAUTION:** A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

 **WARNING:** A WARNING indicates a potential for property damage, personal injury, or death.

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Revision history

Table 1. Document revision history

Part number	Release date	Description of changes
H04504	May 2025	Initial release.

Introduction

This deployment guide covers Dell Technologies PowerSwitch configuration using SmartFabric Manager for SONiC (SFM) for single rack or multirack greenfield PowerStore nonvolatile memory express (NVMe)/TCP storage deployments.

SFM automates the configuration of the leaf-spine fabric that is used in this deployment. The leaf-spine fabric uses Virtual Extensible LAN (VXLAN), which enables a virtual Layer 2 overlay network to run over a physical Layer 3 underlay network. Use of a VXLAN leaf-spine fabric is a best practice in the data center.

Topics:

- [Enterprise SONiC Distribution by Dell Technologies](#)
- [SmartFabric Manager for SONiC](#)
- [PowerStore](#)
- [Typographical conventions](#)
- [We value your feedback](#)

Enterprise SONiC Distribution by Dell Technologies

SONiC is an open-source, Linux-based network operating system that runs on switches from multiple vendors and ASICs. Enterprise SONiC Distribution by Dell Technologies is a hardened and validated version of SONiC that is supported on Dell PowerSwitch platforms. Dell Enterprise SONiC supports an intuitive command-line interface, object-based administration through a REST interface, Google's gRPC Network Management Interface (gNMI), and simplified configuration using SmartFabric Manager.

SmartFabric Manager for SONiC

SmartFabric Manager for SONiC (SFM) simplifies network configuration and maintenance by automating network deployment and providing an easy-to-use web user interface.

SFM is a virtual machine that runs on VMware ESXi, provided in OVA format, or on Linux KVM, provided in QCOW2 format. SFM configures Dell PowerSwitch systems using REST API commands.

The PowerStore deployment in this guide uses a Border Gateway Protocol (BGP) Ethernet VPN (EVPN) with VXLAN leaf-spine fabric. VXLAN enables a virtual Layer 2 overlay network to run over a physical Layer 3 underlay network. Use of a VXLAN leaf-spine fabric is a best practice in the data center. Manually configuring the switches for this fabric would require hundreds of CLI commands. SFM automates the fabric creation, making it fast, easy, and with a much lower probability of errors than configuring the fabric manually.

SFM comes with a 180-day free trial for up to 10 switches. If your deployment uses more than 10 switches in leaf or spine roles, you must buy a license before proceeding. See the *SmartFabric Manager for SONiC User Guide* for more information about licensing.

PowerStore

Built for NVMe technology, PowerStore uses a dual-node appliance architecture that features an active/active controller configuration where both nodes service I/O simultaneously. PowerStore can scale up with single-drive granularity to over 1 PB raw storage capacity per appliance. PowerStore can also scale out to four appliances per cluster.

Typographical conventions

Type	Use
Monospace text	CLI examples
Bold monospace text	• CLI examples that wrap the page • Highlighted information in CLI output
<i>Italic monospace text</i>	Variables in CLI examples
Bold text	• UI fields • Information that is entered in the UI

We value your feedback

Dell Technologies and the authors of this document welcome your feedback on the solution and the solution documentation. Contact the Dell Technologies Solutions team by [email](#).

Authors: Dell Technologies Network Solutions Engineering, Technical Marketing, and Content Engineering and Translation teams

Hardware used in this guide

This section briefly describes the hardware that is used in the deployment example covered in this guide. Validated components contains a complete list of all hardware and software used. Supported Dell PowerSwitch systems provides a complete list of switches that are supported for this deployment.

Topics:

- [Dell PowerSwitch S5248F-ON](#)
- [Dell PowerSwitch S5232F-ON](#)
- [Dell PowerSwitch S3248T-ON](#)
- [Dell PowerStore enterprise storage appliance](#)
- [Dell PowerEdge server](#)

Dell PowerSwitch S5248F-ON

The Dell PowerSwitch S5248F-ON is a 1-Rack Unit (1U), multilayer switch with 48x 25 GbE, 4x 100 GbE, and 2x 200 GbE ports. This guide uses two S5248F-ON switches in each rack as leaf switches.



Figure 1. Dell PowerSwitch S5248F-ON

Dell PowerSwitch S5232F-ON

The Dell PowerSwitch S5232F-ON is a 1U, multilayer switch with 32x 100 GbE QSFP28 ports and 2x 10 GbE SFP+ ports. This guide uses two S5232F-ON switches as spine switches.



Figure 2. Dell PowerSwitch S5232F-ON

Dell PowerSwitch S3248T-ON

The Dell PowerSwitch S3248T-ON is a 1U, multilayer switch with 48x 1 GbE BASE-T ports, 4x 10 GbE SFP+ ports, and 2x 100 GbE QSFP28 ports. This guide uses one S3248T-ON switch in each rack for out-of-band (OOB) management traffic.



Figure 3. Dell PowerSwitch S3248T-ON

NOTE: Other Dell and third-party switches with 1 GbE BASE-T ports and 10 GbE or faster uplink ports can also be used for OOB management traffic.

Dell PowerStore enterprise storage appliance

This guide uses two PowerStore appliances with 25 GbE SFP28 ports for NVMe/TCP storage and cluster communication. See the [Dell Technologies PowerStore](#) web page for PowerStore appliance models and specification sheets.



Figure 4. Dell PowerStore enterprise storage appliance

Dell PowerEdge server

This guide uses two PowerEdge servers with 2x 25 GbE SFP28 ports as NVMe/TCP storage clients.



Figure 5. Dell PowerEdge R670 server

Topology

The topology for this deployment is divided into two parts:

1. Production, or in-band
2. Out-of-band (OOB) management

The production topology contains redundant components and is used for all mission-critical and end-user network traffic. The OOB management network is an isolated network for hardware and PowerStore appliance management.

Topics:

- [Production topology](#)
- [Production topology connection details](#)
- [OOB management topology](#)
- [OOB management connection details](#)

Production topology

The production topology that is used in this guide is shown in the figure below.

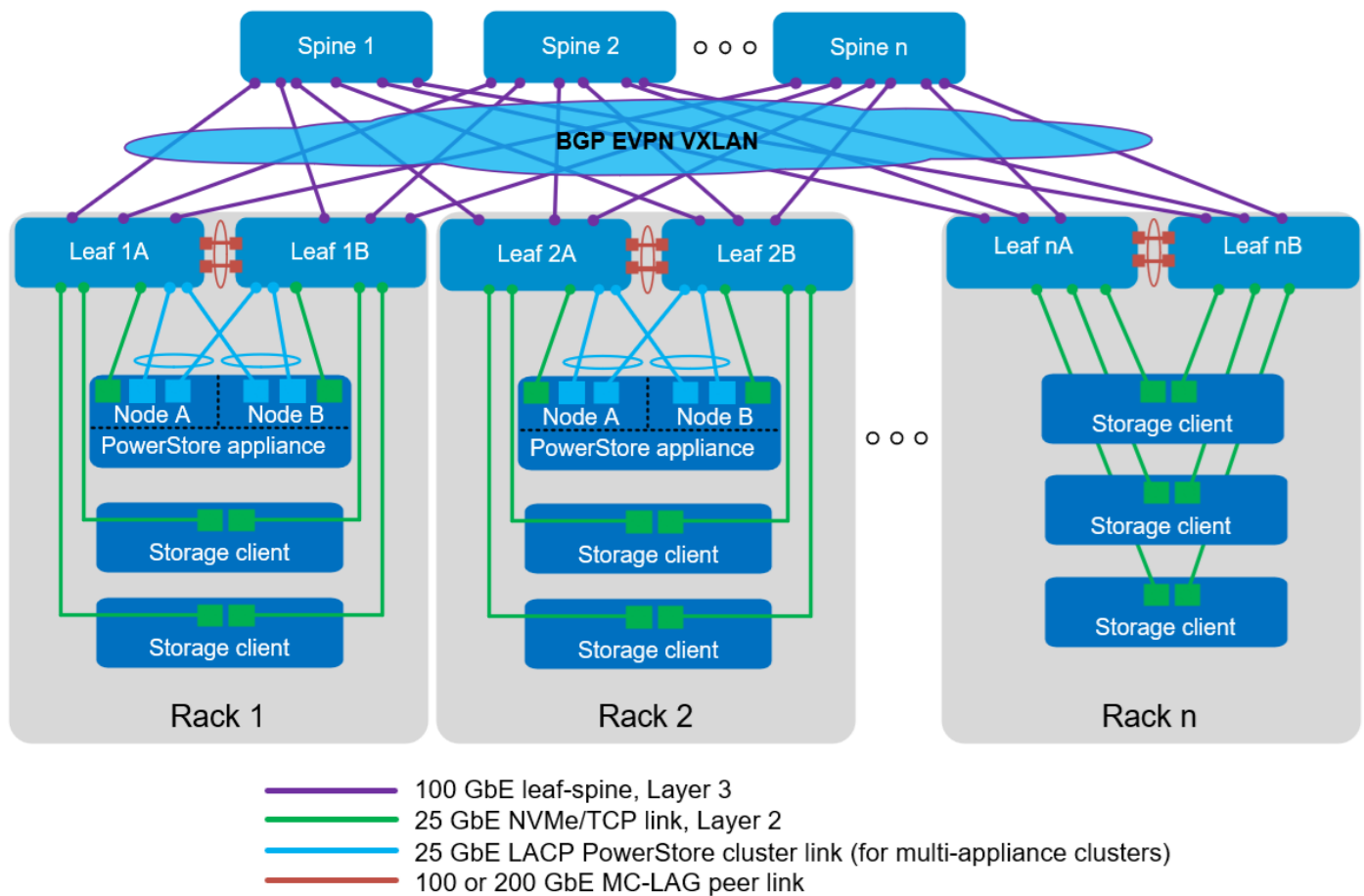


Figure 6. Leaf-spine production topology with PowerStore appliance cluster

The production topology uses a BGP EVPN with VXLAN leaf-spine fabric for performance and scalability.

VXLAN is used to stretch Layer 2 networks across the Layer 3 leaf-spine fabric. Equal-Cost Multi-Path routing (ECMP) is enabled on the switches to use all available bandwidth on the leaf-spine connections. SFM automates the configuration of this VXLAN fabric.

NOTE: See [BGP EVPN with VXLAN overview](#) in the appendix for more information.

Two leaf switches are used in each rack for redundancy and performance. A multichassis link aggregation group (MC-LAG) peer link connects each pair of leaf switches. Every leaf switch is connected to every spine switch to form a leaf-spine topology.

PowerStore appliances that are in the same PowerStore cluster can be in the same rack or in different racks as shown in the figure above. PowerStore cluster network connections, which are shown in blue, use Link Aggregation Groups (LAGs) with Link Aggregation Control Protocol (LACP). If there is only one PowerStore appliance, the PowerStore cluster links shown are not required.

The PowerStore appliance and storage client NVMe/TCP connections, which are shown in green, do not use LAGs in this deployment. There should be at least one NVMe/TCP connection from each PowerStore appliance node and storage client to each leaf switch in the rack for redundancy.

Storage clients can be any PowerEdge or third-party server with at least two 25 GbE or faster NICs. The client operating system must support NVMe/TCP.

NOTE: The example that is detailed in this guide uses two spine switches and the equipment that is shown in Racks 1 and 2 in the figure above. The figure shows how the topology can be scaled out to include additional switches and racks.

Production topology connection details

Production network connection details for this deployment example are shown in the figure below.

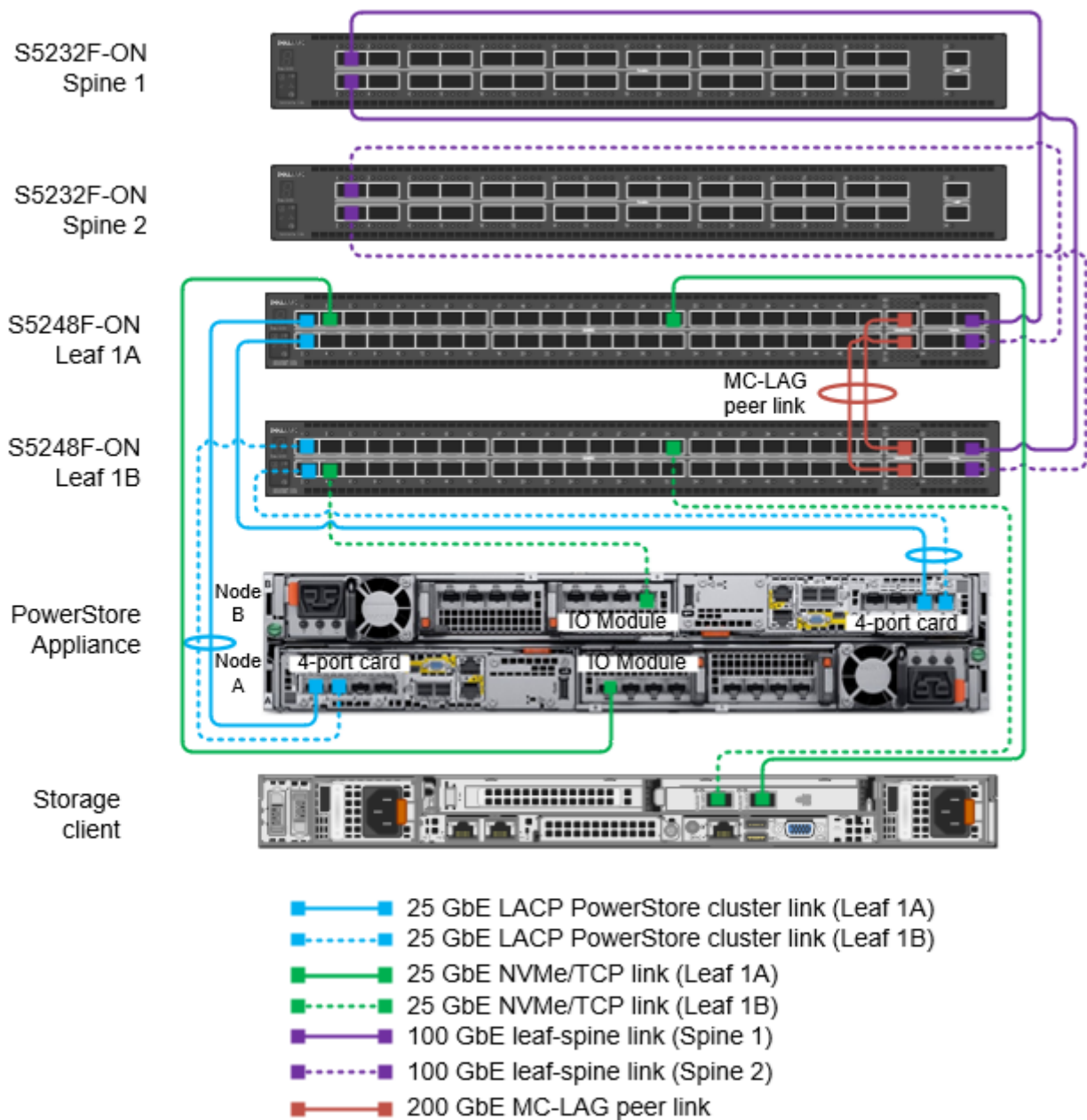


Figure 7. Production network connection details

NOTE: Other PowerStore appliances, leaf switches, and spine switches that are not shown are connected in the same manner. Port speeds and NIC locations vary depending on the equipment used.

Each leaf switch has one connection to each spine switch. For this deployment example, all leaf-spine connections are 100 GbE.

When cabling from the PowerStore appliance node ports to the leaf switches, the corresponding ports on Node A and Node B should connect to opposite leaf switches.

Ports 0 and 1 on the PowerStore 4-port cards are preconfigured for LACP and are used for the cluster network connections, shown in blue. If you do not have a multi-appliance cluster, these connections are not required.

NOTE: Ports 0 and 1 on the 4-port card can also be used for other traffic. For more information, see [Dell PowerStore: Introduction to the Platform](#). If deploying a PowerStore 500T appliance, ports 2 and 3 on the 4-port card are reserved for connectivity to NVMe expansion enclosures. An expansion enclosure is not used in this example.

For the MC-LAG peer link connections, 100 GbE or faster ports are typically used. Dell Technologies recommends using at least two physical ports on each leaf switch for the peer link. If 25 GbE ports are used for the peer link, Dell Technologies recommends using at least four connections on each leaf for sufficient bandwidth. Multiple peer link connections provide

redundancy and additional bandwidth if a switch is taken offline or if there is a failure. The peer link is used for data traffic only if there is a link failure that requires the peer link to reach the destination.

NOTE: On S5248F-ON switches, two QSFP28-DD double density ports (2 x 100 GbE interfaces per physical port), are available. They are used in this example to create a 400 GbE MC-LAG peer link. This connection requires QSFP28-DD DAC cables or optics.

OOB management topology

The out-of-band (OOB) management network is an isolated network for remote management of PowerStore appliances and PowerSwitch systems. It can also include other devices using their dedicated management ports.

One S3248T-ON switch is typically installed in each rack for OOB management network connections, as shown in the following figure. Other Dell and third-party switches with 1 GbE BASE-T ports can also be used for OOB management traffic.

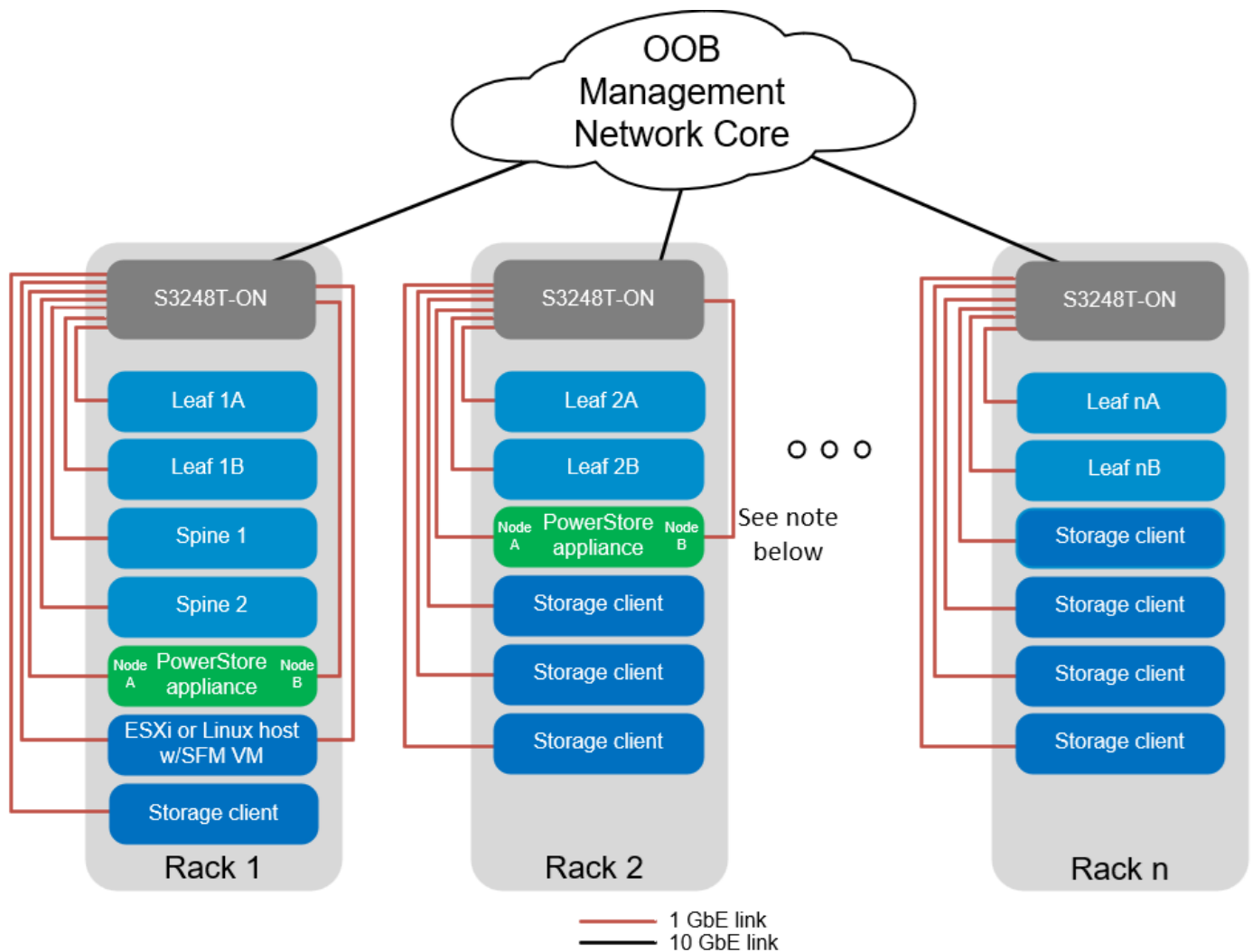


Figure 8. OOB management network topology

NOTE: If multiple PowerStore appliances are deployed in a cluster, all appliance management ports in the cluster must be untagged in the same Layer 2 VLAN. If PowerStore appliances are in different racks, you can keep them in the same VLAN by using Layer 2 uplinks to the OOB management network core switch. If you use Layer 3 uplinks to the OOB management network core switch, connect all PowerStore appliances to the same physical OOB management switch, even if the appliances are in different racks.

Four 10 GbE SFP+ ports are available on each S3248T-ON. One or more of these ports are used as uplinks to the OOB management network core.

The 1 GbE 1000BASE-T ports on each S3248T-ON are connected downstream to hardware management ports on each device in the rack. Those ports include the PowerSwitch and PowerStore management ports and the PowerEdge server iDRAC ports. Dedicated management ports on other devices in the rack can also be connected to the OOB management switches.

SFM uses the OOB management network to configure and manage the leaf and spine switches that form the production topology. The ESXi or Linux host with the SFM VM must be accessible on this network. For this reason, the host that contains SFM requires a 1 GbE NIC connected to the OOB management network in addition to its iDRAC port.

NOTE: SFM does not manage the OOB management switches.

The OOB management network is used for initial PowerStore configuration, and to access PowerStore Manager. It is also used to discover other PowerStore appliances that will be added to a cluster. All switch ports connected to PowerStore management ports are untagged (also known as access ports) in the same VLAN as shown in [Configure OOB management switches](#).

NOTE: For reference, devices on the OOB Management network in this guide use the 100.67.0.0/16 IP address range. All IP addresses that are used in this guide are examples only.

OOB management connection details

All device management ports are connected to the OOB management switch in each rack as shown below.

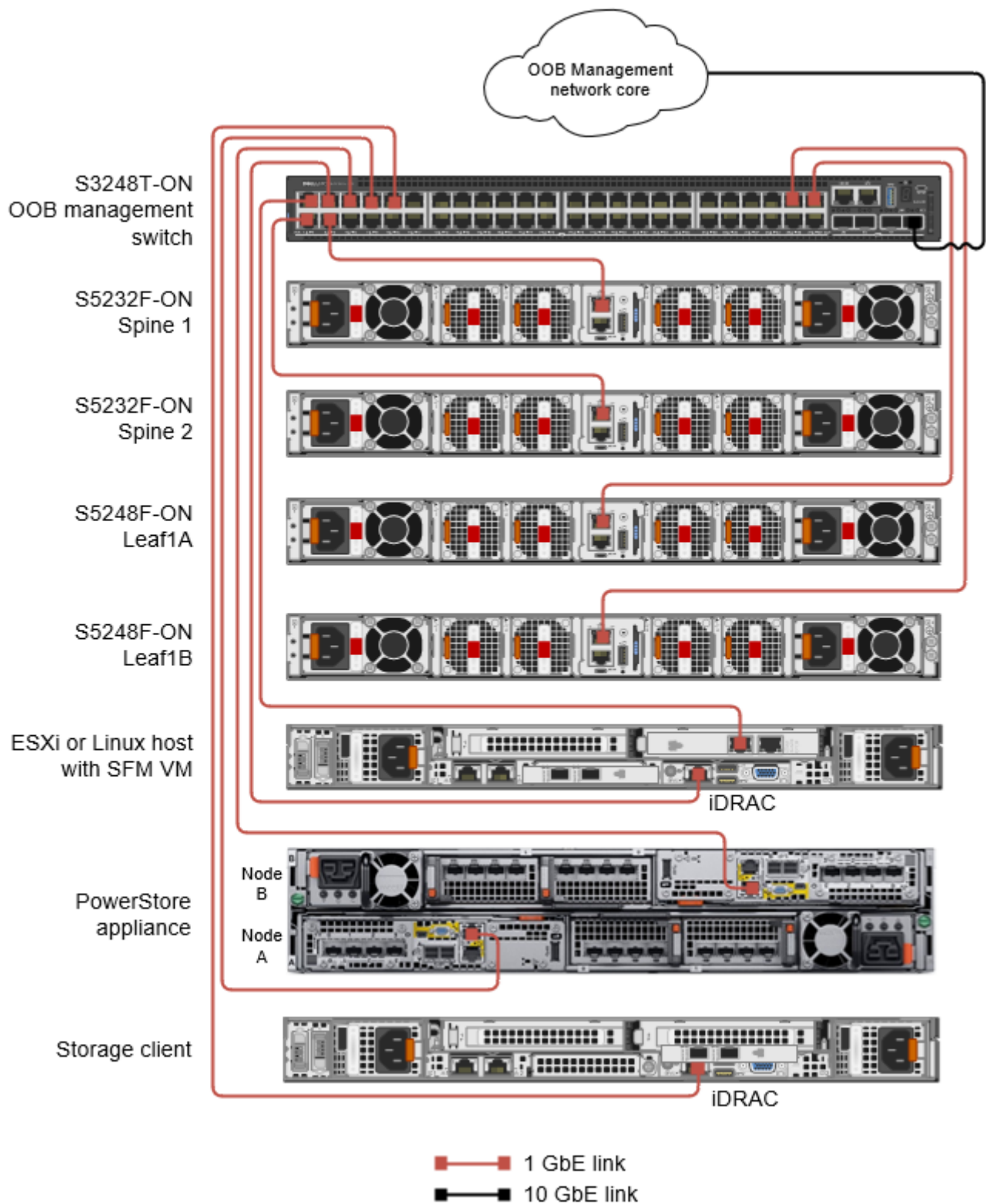


Figure 9. OOB management network connection details

NOTE: Other leaf switches, spine switches, PowerStore appliances, and hosts that are not shown are connected in the same manner. Port locations will vary depending on the equipment used.

The ESXi or Linux host with the SFM VM has two connections to the OOB management network: the 1 GbE iDRAC and an additional 1 GbE connection for SFM communication with the PowerSwitch leaf and spine systems.

Each PowerStore appliance has two connections to the OOB management network, one for each node.

Deployment Planning

This chapter covers items to consider before deploying SFM for PowerStore.

Topics:

- [Minimum networking requirements](#)
- [Supported Dell PowerSwitch systems](#)
- [PowerStore VLANs and IP addresses](#)

Minimum networking requirements

NOTE: In this guide, “single rack” means topologies with only two leaf switches. “Multirack” means topologies with four or more leaf switches.

The minimum networking requirements for PowerStore deployments with SFM include the following:

- Single rack deployments require two supported Dell PowerSwitch leaf switches of the same model.
- Multirack deployments require two supported Dell PowerSwitch leaf switches of the same model per rack, plus at least two supported Dell PowerSwitch spine switches.
- All Dell PowerSwitch leaf and spine switches in the fabric must be running the same version of Dell Enterprise SONiC. (Temporary exceptions are allowed during SONiC OS upgrades.)
- Dell Enterprise SONiC 4.4.1 or later with an Enterprise Standard or Enterprise Premium bundle is required on all leaf and spine switches in the fabric.
- At least one 1 GbE BASE-T switch is required for OOB management connections. Dell Technologies recommends using one PowerSwitch S3248T-ON per rack. If Dell PowerSwitches are used for OOB management, Dell Technologies recommends Enterprise SONiC 4.4.1 or later with the Lite bundle.
- Each PowerStore appliance requires at least one 25 GbE SFP28 or faster connection to each leaf switch in the rack for NVMe/TCP storage traffic. As a best practice, do not use a link aggregation group (LAG) for NVMe/TCP connections.
- If multiple PowerStore appliances are used in a cluster, each PowerStore appliance requires two additional 25 GbE or faster connections to each leaf switch in the rack for cluster traffic. These connections must be in a LAG with Link Aggregation Control Protocol (LACP) enabled.
- MC-LAG peer links between leaf switch pairs require at least two connections for fault tolerance. Dell Technologies recommends using 100 GbE or faster ports for these connections. If 25 GbE ports are used for the peer link, Dell Technologies recommends using at least four connections for sufficient bandwidth.
- One 1 GbE (1000BASE-T) connection is required for each OOB management port on leaf and spine switches.
- One 1 GbE (1000BASE-T) connection is required for each of the two OOB management ports on PowerStore appliances.
- If PowerEdge servers are used as storage clients, one 1 GbE (1000BASE-T) connection is recommended for each server’s iDRAC port.
- One server running ESXi 7.0 or later or Linux KVM is required to host the SFM VM. The SFM VM requires at least 32 GB RAM, 120 GB free disk space, and 16 virtual CPUs. In addition to the server’s iDRAC 1 GbE (1000BASE-T) connection, the host server must have a network connection that allows the VM to reach the OOB management network. SFM uses this connection to communicate with the leaf and spine switches.

Supported Dell PowerSwitch systems

Dell PowerSwitch systems that are supported as leaf or spine switches with SFM for PowerStore NVMe/TCP in VXLAN fabrics are listed in the table below.

Table 2. Supported switches in leaf or spine roles in VXLAN fabrics

Dell PowerSwitch model	Typical role ¹	Notes
S5212F-ON S5224F-ON S5248F-ON S5296F-ON	25 GbE leaf	None
S5232F-ON	100 GbE leaf or spine	None
S5448F-ON	100 GbE leaf	None
Z9264F-ON	100 GbE spine	Z9264F-ON is not supported as a leaf.
Z9332F-ON	400 GbE spine	Z9332F-ON is not supported as a leaf.
Z9432F-ON	400 GbE spine	None
Z9664F-ON	400 GbE spine	Z9664F-ON is not supported as a leaf.

¹Any Dell PowerSwitch in the table above can be used as a leaf or a spine in a VXLAN fabric except where noted. Breakout cables and other supported port speeds can be used if needed.

For OOB management, Dell Technologies recommends one PowerSwitch S3248T-ON per rack. Other Dell or third-party switches with 1000BASE-T ports can also be used for the OOB management network.

PowerStore VLANs and IP addresses

The VLAN IDs used in this guide match those used in the [Dell PowerStore T and Q Models Networking Guide for Initial Deployment](#) and [Dell PowerStore T and Q Networking Guide for Storage Services](#). Use the worksheets provided in those guides to plan your VLANs and IP addresses.

Table 3. VLANs and IP addresses that are used in this guide

VLAN ID	Description	Where used	Switch ports	IP address block
100	PowerStore management network	OOB Management switch	Untagged (Access)	100.67.0.0/16
1	PowerStore cluster network	SFM leaf-spine fabric	Untagged (Access)	IPv6 link local
300	PowerStore NVMe/TCP storage	SFM leaf-spine fabric	Tagged (Trunk)	192.168.3.0/24

 **NOTE:** The VLAN IDs and IP addresses used in this guide are examples only and can be changed as needed. In an SFM fabric, VLAN IDs 1-3446, 3939, and 3961-3966 are available for use. SFM reserves the other VLAN IDs.

Configure OOB management switches

This section shows an OOB management switch configuration example. It uses a Dell PowerSwitch S3248T-ON running Dell Enterprise SONiC 4.4.1 Lite. The switch starts at its factory default configuration.

NOTE: OOB management switches can be Dell PowerSwitch systems or third-party switches with 1 GbE BASE-T ports. If you are using an operating system other than Dell Enterprise SONiC, refer to your operating system's documentation for specific commands.

1. Connect to the switch using the console serial port.
2. Log in to the CLI using the default username `admin` and password `YourPaSsWoRd`.
3. Change the admin password to a secure one when prompted.
4. Run the following commands to use standard interface names and to set the hostname.

```
admin@sonic:~$ sonic-cli
sonic# configure terminal
sonic(config)# interface-naming standard

Broadcast message: Interface naming mode has changed. Users running 'sonic-cli' are
required to restart your session.

sonic(config)# hostname S3248T-OOB
Broadcast message: Hostname has been changed from 'sonic' to 'S3248T-OOB'. Users
running 'sonic-cli' are suggested to restart your session.
```

5. Exit to apply the changes and return to the SONiC CLI.

```
sonic(config)# exit
sonic# exit
admin@sonic:~$ sonic-cli
S3248T-OOB#
```

6. Enabling a spanning tree protocol is a best practice. Rapid per-VLAN spanning tree is enabled in this example.

```
S3248T-OOB# configure
S3248T-OOB(config)# spanning-tree mode rapid-pvst
```

7. Create the OOB management VLAN.

```
S3248T-OOB(config)# interface Vlan 100
```

NOTE: VLAN ID 100 is an example. Any available VLAN ID can be used.

8. Configure all downlink ports that will be connected to PowerStore management ports, PowerEdge server iDRACs, and other devices in the rack if applicable. Put them in the same VLAN as access (untagged) ports and bring the ports up administratively.

```
S3248T-OOB(config)# interface range Eth 1/1-1/48
%Info: Configuring only existing interfaces in range
S3248T-OOB(config-if-range-eth**)# switchport access Vlan 100
S3248T-OOB(config-if-range-eth**)# no shutdown
```

9. Configure the uplink to the OOB management network core switch. Details are not shown here as the uplink can be configured using different Layer 2 or Layer 3 options. If the uplink is in Layer 3 mode, all PowerStore appliances in a cluster must be connected to the same OOB management switch (see note below).

NOTE: If multiple PowerStore appliances are deployed in a cluster, all appliance management ports in the cluster need to be untagged in the same Layer 2 VLAN. If PowerStore appliances are in different racks, you can keep them in the same VLAN by using Layer 2 uplinks to the OOB management network core switch. If you use Layer 3 uplinks to the OOB management network core, connect all PowerStore appliances to the same physical OOB management switch, even if the appliances are in different racks.

10. Save the configuration.

```
S3248T-OOB(config-if-range-eth**)# end
S3248T-OOB# write memory
```

Use SFM to create the leaf-spine fabric

This chapter shows how to use SFM to create the VXLAN fabric for a PowerStore deployment. For other SFM features and additional information, see the *SmartFabric Manager for SONiC User Guide*.

Topics:

- [Prepare leaf and spine switches for SFM](#)
- [Add leaf and spine switches to SFM inventory](#)
- [Create the fabric](#)
- [Create a nondefault tenant](#)
- [Create the networks](#)
- [Create downlinks to PowerStore appliances and storage clients](#)

Prepare leaf and spine switches for SFM

Set the admin password and configure the management interface

For each Dell PowerSwitch that will be a leaf or a spine in the fabric, do the following:

1. Connect to the switch using the console serial port.
2. Log in to the CLI as **admin**.
3. Change the admin password from the default, **YourPaSsWoRd**, to a secure password during the first login when prompted.
4. Configure the OOB management interface (Management 0). SFM uses this interface to communicate with the leaf and spine switches. For SFM, Dell Technologies recommends using a static IP address instead of DHCP on Management 0 to ensure that it does not change. If routing is used on your OOB management network, also specify the gateway address as shown in the configuration example below.

```
admin@sonic:~$ sonic-cli
sonic# configure
sonic(config)# interface Management 0
sonic(config-if-Management0)# ip address 100.67.112.13/24 gwaddr 100.67.112.254
sonic(config-if-Management0)# end
sonic# write memory
```

The commands above are the only ones that are required at the switch CLI when using SFM. The IP addresses used are examples only.

NOTE: If any leaf or spine switches have previously been configured (other than changing the admin password and configuring Management 0), run the `write erase` command and reboot each applicable switch. The admin password and Management 0 interface configuration will be retained. All other configuration settings on the switch will be reset to the factory defaults.

Connect the switches and SFM host

Make the switch and SFM host connections as shown in [Production topology connection details](#) and [OOB management network connection details](#). The PowerStore appliances and storage clients do not have to be connected now.

Deploy the SFM virtual machine to a VMware ESXi or Linux host as shown in the *SmartFabric Manager for SONiC User Guide*. SFM is available in OVA format for ESXi or in QCOW2 format for Linux KVM.

The host for the SFM VM must have network connectivity to the OOB management network for SFM to communicate with the leaf and spine switches. Example network connections are shown in [OOB management network connection details](#).

Add leaf and spine switches to SFM inventory

1. Connect to the SFM UI in a browser and log in.
2. In the left pane, select **Inventory > Switches**.
3. In the right pane, click the **Discovery Requests** tab and click **+DISCOVER**. The Discover Switches dialog box opens.
 - a. Next to **Discovery Mode**, select either **Range** or **List**. **Range** is selected in this example.
 - b. Enter the OOB Management addresses of the leaf and spine switches as a range or a comma-separated list.
 - c. (Optional) Enter a name to use as the **Discovery ID**.
 - d. Leave the **Port** set to **443**.
 - e. Enter the **Username** and **Password** for the switches. All switches in one discovery instance must use the same credentials.

Discover Switches

Discovery Mode

Range

IPv4 Example: Start IP Address 1.1.1.1 End IP Address 1.1.1.10

Start IP Address

100.67.112.13

IPv4 Address

End IP Address

100.67.112.18

IPv4 Address

Discovery Id
(Optional)

PS leafs and spines

Ex. Discovery-Networks

Port

443

Username

admin

Password

.....

CANCEL

DISCOVER

Figure 10. Discover switches by IP address range

4. Click **DISCOVER**. The discovery process completes in less than two minutes for this example. The switches are listed on the **Discovered Switches** tab of the **Inventory** page in SFM.

Inventory

Switch Inventory

Discovery Requests

Discovered Switches

REFRESH

DELETE

ADD TO INVENTORY

☐	Switch Name	Discovery Id	IP Address
☐	sonic	PS leafs and spines	100.67.112.14
☐	sonic	PS leafs and spines	100.67.112.15
☐	sonic	PS leafs and spines	100.67.112.16
☐	sonic	PS leafs and spines	100.67.112.13
☐	sonic	PS leafs and spines	100.67.112.18
☐	sonic	PS leafs and spines	100.67.112.17

Discovered Switches per page

10

1 - 6 of 6 Nodes

Figure 11. Discovered switches

5. Check the box next to all switches that will be in the leaf-spine fabric and then click **ADD TO INVENTORY**.
6. In the **Add Switches to Inventory** dialog box, enter a **Name** and **Role** (select **Leaf** or **Spine**) for each switch. Leave the **Monitor Only** toggle button set to the off position as shown.

Add Switches to Inventory

IP Address	Name	Role	MonitorOnly
100.67.112.14	leaf1b	Leaf v	☐
100.67.112.13	leaf1a	Leaf v	☐
100.67.112.15	leaf2a	Leaf v	☐
100.67.112.18	spine2	Spine v	☐
100.67.112.16	leaf2b	Leaf v	☐
100.67.112.17	spine1	Spine v	☐

Figure 12. Add switches to inventory

- Click **SUBMIT** to add the switches to the SFM inventory.
- The switches are listed on the **Switch Inventory** tab with a status of **Offline**. The status takes less than two minutes to change to **Online** for this example.
- Click **REFRESH** to update the **Status** column as needed.

Inventory

Switch Inventory

Discovery Requests

Discovered Switches

REFRESH

+ ADD SWITCH

EDIT

DELETE

CLAIM OWNERSHIP

RUN CLI COMMANDS

+ CREATE FABRIC

ADD TO FABRIC

☐		Name	Role	IP Address	Status	Inventory
☐	>>	leaf1a	Leaf	100.67.112.13	Online	✓ Added
☐	>>	leaf1b	Leaf	100.67.112.14	Online	✓ Added
☐	>>	leaf2a	Leaf	100.67.112.15	Online	✓ Added
☐	>>	leaf2b	Leaf	100.67.112.16	Online	✓ Added
☐	>>	spine1	Spine	100.67.112.17	Online	✓ Added
☐	>>	spine2	Spine	100.67.112.18	Online	✓ Added

Nodes per page
10
1 - 6 of 6 Nodes

Figure 13. Switch Inventory tab

Create the fabric

- On the **Switch Inventory** tab, check the box next to each switch that will be part of the fabric.

Inventory

Switch Inventory

Discovery Requests

Discovered Switches



REFRESH

+ ADD SWITCH

EDIT

× DELETE



CLAIM OWNERSHIP



RUN CLI COMMANDS

+ CREATE FABRIC



ADD TO FABRIC

☒		Name	Role	IP Address	Status	Inventory
☒	>>	leaf1a	Leaf	100.67.112.13	Online	✓ Added
☒	>>	leaf1b	Leaf	100.67.112.14	Online	✓ Added
☒	>>	leaf2a	Leaf	100.67.112.15	Online	✓ Added
☒	>>	leaf2b	Leaf	100.67.112.16	Online	✓ Added
☒	>>	spine1	Spine	100.67.112.17	Online	✓ Added
☒	>>	spine2	Spine	100.67.112.18	Online	✓ Added
☒	6					Nodes per page 10 1 - 6 of 6 Nodes

Figure 14. Fabric switches selected

- Click **+CREATE FABRIC**. The **Create Fabric** dialog box opens.

Fabric Name	PowerStore Fabric	Leaf ASN	65011 1 - 4294967295
Fabric Technology	VxLan	Spine ASN	65012 1 - 4294967295
Site	1	Super Spine ASN	65013 1 - 4294967295
STP Edge	Enable	Border Leaf ASN	65014 1 - 4294967295
LACP Fallback mode	SwitchConfigured	BGP CIDR	172.16.128.0/24 Ex: 10.0.0.0/24
ROCE	☐	VTEP CIDR	172.30.0.0/24 Ex: 10.0.0.0/24
Cut Through Switching	☐		
ARS	☐		

Layer 3 Only
 • Enabling RoCE Will Reboot Switches
 • Enabling ARS will enable Dynamic Load Balancing on ISL links

> Border Leaf (0)

> Super Spine (0)

> Spine (2)

> Leaf (4)

CONFIGURE MC-LAG PEER LINK

	Name	Role	IP Address	MAC Address	MC-LAG Peer Links
☐	leaf1a	Leaf	100.67.112.13	54:bf:64:e6:dd:c3	VIEW
☐	leaf1b	Leaf	100.67.112.14	3c:2c:30:10:35:02	VIEW
☐	leaf2a	Leaf	100.67.112.15	3c:2c:30:89:45:82	VIEW
☐	leaf2b	Leaf	100.67.112.16	3c:2c:30:22:79:82	VIEW

Nodes per page 10 1 - 4 of 4 Nodes

Figure 15. Create fabric dialog box

- In the **Create Fabric** dialog box, do the following:
 - Enter a **Fabric Name**.
 - Leave the **Fabric Technology** set to the default, **VXLAN**.
 - Enter a **Site** name.
 - Set **STP Edge** to **Enable** (recommended). This enables the Spanning Tree Protocol (STP), specifically Rapid Per-VLAN Spanning Tree (RPVST), on the fabric. It also configures downlink ports from leaf switches as STP edge ports. Enabling STP is a best practice against loops in case there is a misconfigured fabric. In a properly configured VXLAN fabric, all paths in the fabric are active and no connections are blocked by STP. If you do not want STP configured like this, you can set **STP Edge** to **Switch Configured**. This allows you to make custom STP settings after fabric creation using the CLI at the console of each switch. Optionally, you can select **Disable**.
 - Set **LACP Fallback** mode to either **Switch Configured** (recommended) or **Fallback None**. LACP fallback is not used in PowerStore deployments. Setting it to **Switch Configured** allows you to configure it after fabric creation using the switch CLI if needed for other devices. Optionally, you can set it to **Fallback None** if you do not need the LACP fallback feature.
- Keep **ROCE**, **Cut Through Switching**, and **ARS** set to **off** for this VXLAN fabric. The toggle buttons for **ROCE**, **Cut Through Switching**, and **ARS** only apply to a traditional L3 fabric.
- In the right column, notice that the Leaf and Spine **ASN** and BGP and VTEP **CIDR** fields are prepopulated with default settings. Dell Technologies recommends leaving the ASNs and IP address blocks at their defaults unless the settings conflict with your existing environment.

NOTE: Switches in the Super Spine and Border Leaf roles are not used in this deployment.

- In the table at the bottom of the window, click the **>** next to **Leaf**. The leaf switches are listed. The **MC-LAG Peer Links** are automatically detected if they are properly connected when adding the switches to SFM. You can view the detected peer link ports by clicking **VIEW** in the table next to the applicable leaf switch. Optionally, the peer link ports can be edited by selecting a leaf and clicking **CONFIGURE MC-LAG PEER LINK**.
- In the table at the bottom of the window, click the **>** next to **Spine**. Ensure that all spine switches are listed if applicable.
- Click **CREATE FABRIC**. A green **Success** message displays at the top of the page. You might need to click **REFRESH** to see the fabric in the list.

Create Fabric [PowerStore Fabric]: Success

Managed Fabrics

REFRESH EDIT DELETE

	Fabric Name	No. of Leaves	No. of Spines	No. of SuperSpines	No. of BorderLeaves	Fabric Type	Site	Action
☐	PowerStore Fabric	4	2	0	0	VxLan	1	MANAGE

Fabrics per page 10 1 - 1 of 1 Fabrics

Figure 16. Fabric created

- To see the fabric layout, click **MANAGE**. The **Fabric: Fabric Name** page opens.
- Click the **Topology** tab. The leaf-spine topology of the current fabric displays. Hovering your mouse pointer over a link provides the port numbers of the connection. Hovering your mouse pointer over a switch provides the switch details, as shown in the figure below.

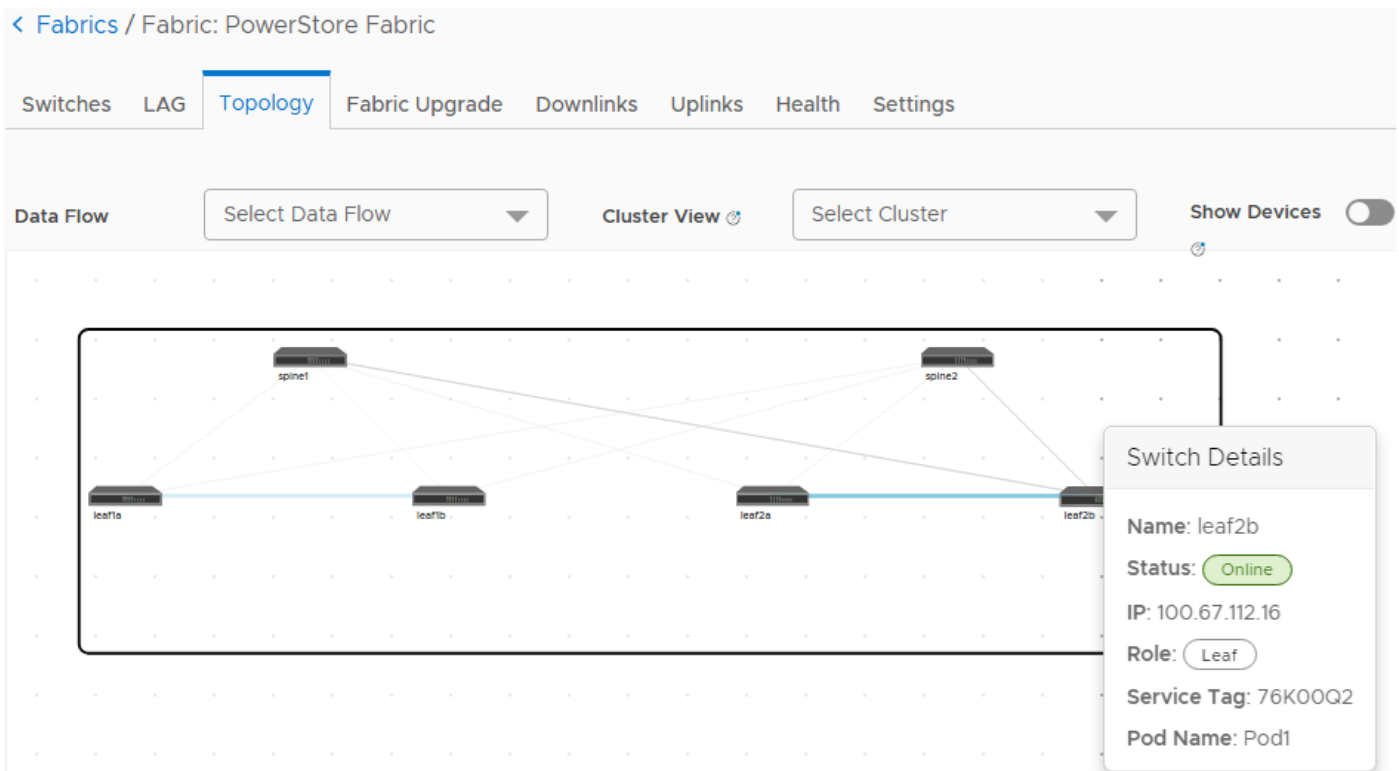


Figure 17. Fabric topology view

Create a nondefault tenant

Virtual Routing and Forwarding (VRF) allows multiple instances of independent routing tables to co-exist within the same router simultaneously. SFM refers to VRFs as tenants.

Using at least one nondefault tenant is a best practice because it enables symmetric integrated bridging and routing (IRB) with VXLAN. In symmetric IRB, both the ingress and egress VTEPs route packets to their destination. The alternative is asymmetric IRB, which does not handle routing as efficiently as symmetric IRB.

To create a nondefault tenant for the fabric, do the following:

- 1. In the SFM left pane, click **Resources**. The **Tenants** tab is displayed. The default tenant in SFM is named **Global**.
- 2. Click **+ADD**. The **Add Tenant** dialog box displays.

Add Tenant

Name

PS-tenant

Description

Enter Description

CANCEL

SUBMIT

Figure 18. Add tenant

- 3. Enter a **Name** for the tenant. Do not use a space in the tenant's name.
- 4. Click **SUBMIT**. A green **Success** message displays, and the new tenant is listed.

Resources

Tenants

Networks

Route Policy

Create Tenant [PS-tenant]: Success

REFRESH

+ ADD

X DELETE

	Name	Description	Status
☐	Global	Global	
☐	PS-tenant		

Tenants per page 10 1 - 2 of 2 Tenants

Figure 19. Nondefault tenant created

NOTE: This nondefault tenant is used for all networks that are used by the PowerStore appliances and storage clients. It can also be used for any other networks in the fabric.

Create the networks

Create the NVMe/TCP storage network

A network for PowerStore NVMe/TCP storage traffic is created in this section. The values shown in the figure below are from [VLANs and IP addresses that are used in this guide](#).

1. In the SFM left pane, select **Resources**.
2. On the **Resources** page, click the **Networks** tab.
3. On the **Networks** tab, click **+ADD > VxLAN Network**.
4. In the **Add VxLAN Network** dialog box, enter a **Network Name**, select the nondefault **Tenant** that is created in the previous section, enter an ID in the **Vlan** field.

 **NOTE:** SFM reserves a range of VLAN IDs for internal use. Available VLAN ID numbers are listed below the **Vlan** field on the page.
5. (Optional) Specify an anycast address for the NVMe/TCP network.
This address serves as the gateway address for the network and is only needed if devices on this network need to communicate with other networks.
 - a. Click **+** next to **Anycast Address** to expand the box.
 - b. Enter the anycast address in CIDR format as shown.The completed page looks like the following:

Network Name

NVMe-300

Tenant

PS-tenant x

Vlan

300

1-3446, 3939, 3961-3966

Anycast Address

+

Address1

192.168.3.254/24

(e.g. 1.1.1.254/16)

CANCEL

SUBMIT

Figure 20. Add VxLAN Network dialog box – NVMe network

6. Click **SUBMIT**. A green **Success** message displays, and the NVMe network is listed.

Resources

Tenants

Networks

Route Policy

Create Network [NVMe-300]: Success

REFRESH

+ ADD

EDIT

DELETE

	Network Name	Network Type	Tenant Name	VLAN
>>	NVMe-300	VxLan	PS-tenant	300

Networks per page

10

1 - 1 of 1 Networks

Figure 21. NVMe/TCP network created

You can configure additional networks in the same manner if needed.

Create the PowerStore cluster network

The PowerStore cluster network is only used for deployments with multiple PowerStore appliances that are configured in a cluster. If you do not have a PowerStore cluster, go to [Create downlinks to PowerStore appliances and storage clients](#). The values that are shown in the figure below are from [VLANs and IP addresses that are used in this guide](#).

- In the SFM left pane, select **Resources**.
- On the **Resources** page, click the **Networks** tab.
- On the **Networks** tab, click **+ADD > VxLAN Network**.
- In the **Add VxLAN Network** dialog box, enter a **Network Name**, select the nondefault **Tenant** that is created in the previous section, enter an ID in the **Vlan** field.

NOTE: SFM reserves a range of VLAN IDs for internal use. Available VLAN ID numbers are listed under the VLAN entry line on the page.
- Leave the **Anycast Address** field blank. It is not needed for the PowerStore cluster network. The completed page looks like the following:

Add VxLAN Network

Network Name

cluster-1

Tenant

PS-tenant x

Vlan

1

1-3446, 3939, 3961-3966

Anycast Address

+

CANCEL

SUBMIT

Figure 22. Add VXLAN Network dialog box – cluster network

6. Click **SUBMIT**. A green **Success** message displays, and the cluster network is added to the list.

Resources

Tenants

Networks

Route Policy

Create Network [cluster-1]: Success

REFRESH

+

ADD

EDIT

DELETE

	Network Name	Network Type	Tenant Name	VLAN
>>	cluster-1	VxLan	PS-tenant	1
>>	NVMe-300	VxLan	PS-tenant	300

Networks per page

10

1 - 2 of 2 Networks

Figure 23. Cluster network created

Create downlinks to PowerStore appliances and storage clients

In this section, downlinks are configured from the leaf switches to the PowerStore appliances and storage clients.

Configure NVMe/TCP network downlinks

The NVMe/TCP network downlinks to PowerStore appliances and storage clients are added as follows:

1. In the SFM left pane, select **Fabric**.
2. In the right pane next to the fabric name, click **MANAGE**.

3. On the **Fabric: Fabric Name** page, click the **Downlinks** tab.
4. Click **+ADD**.
5. In the **Create Downlink** dialog box, do the following:
 - a. Enter a **Name** for the downlink.
 - b. Next to **Tenant**, select the nondefault tenant that you created in [Create a nondefault tenant](#).
 - c. Leave **Interface Type** set to physical.
 - d. Select the leaf **Switch** and **Interface** that the downlink is connected to.
 - e. If the downlink is connected to a PowerStore, select the NVMe VLAN next to **Tagged Network** as shown below. If the downlink is connected to a storage client, and the NVMe VLAN ID is also configured on the client interface, select the NVMe VLAN next to **Tagged Network**. If a VLAN ID is not configured on the storage client interface, select the NVMe VLAN next to **Untagged Network**.

Create Downlink

Name
500T-nodeA-NVMe

Tenant
PS-tenant x

Port Configuration

Interface Type
☐ LAG
☒ Physical

Switch
leaf1a x
Eth1/10 x

Network Configuration

+ ADD NETWORK

Untagged Network
Select Untagged Network

Tagged Network
NVMe-300 x

CANCEL

SUBMIT

Figure 24. Create Downlink dialog box

- f. Click **SUBMIT**. A green **Success** message displays, and the new downlink appears in the list

< Fabrics / Fabric: PowerStore Fabric

Switches
LAG
Topology
Fabric Upgrade
Downlinks
Uplinks
Health
Settings

Success: Downlink: 500T-nodeA-NVMe created successfully.

REFRESH
+ ADD
EDIT
DELETE

	Name	Info	Tenant Name	Status
☐ >>	500T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	InProgress

Downlinks per page 10
1 - 1 of 1 Downlinks

Figure 25. Fabric downlinks tab – PowerStore downlink created

When SFM has configured the downlink on the switches, the **Status** changes from **In Progress**, as shown, to **Success**. The status might take some time to change. You can click **REFRESH** as needed to update the status. You do not have to wait for the status to change before configuring the next downlink.

Repeat the above steps for each link from the fabric leaf switches to PowerStore and storage client interfaces that are on the NVMe network.

For this example, there are two PowerStore appliances (named 500T and 1000T) and two storage clients (named SC01 and SC02) with two connections each. When complete, the **Downlinks** page appears as shown in the figure below.

< Fabrics / Fabric: PowerStore Fabric

Switches

LAG

Topology

Fabric Upgrade

Downlinks

Uplinks

Health

Settings

Success: Downlink: SC02-p2-NVMe created successfully.

REFRESH

+ ADD

EDIT

DELETE

	Name	Info	Tenant Name	Status
☐ >>	500T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	500T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	1000T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	1000T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC01-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC01-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC02-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC02-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	InProgress

Downlinks per page

10

1 - 8 of 8 Downlinks

Figure 26. NVMe downlinks configured for PowerStore and storage clients

If your storage clients are connected to the leaf switches and have IP addresses configured on the storage network, they should now be able to ping each other.

Configure cluster network downlinks

The PowerStore cluster network is only needed for deployments with more than one PowerStore appliance. If you only have one PowerStore appliance, go to [Configure PowerStore appliances](#).

Cluster downlinks to PowerStore appliances use LAGs with LACP. To add the downlink LAGs for the PowerStore appliances, do the following

1. In the SFM left pane, select **Fabric**.
2. In the right pane next to the fabric name, click **MANAGE**.
3. On the **Fabric: Fabric Name** page, click the **LAG** tab.
4. Click **+ADD**.
5. In the **Create LAG** dialog box, do the following:
 - a. Enter a **Name** for the LAG.
 - b. Set the **Switch Group** to **MC-LAG Leaf**.
 - c. Next to **Switch**, select the leaf switch pair and interfaces that form that LAG.
 - d. Set the **Admin State** to **Up**, **Port Channel Type** to **MC-LAG**, and **Port Channel Mode** to **LACP**.

Create LAG

Name

500T-nodeA-LAG

Switch Group

☒ MC-LAG Leaf ☐ Single Leaf

Switch

leaf1a - leaf1b x

leaf1a:Eth1/11 x leaf1b:Eth1/11 x

Admin State

☒ Up ☐ Down

Port Channel Type

☒ McLAG ☐ Non McLAG

Port Channel Mode

☒ LACP ☐ Static

CANCEL

SUBMIT

Figure 27. Create LAG dialog box

e. Click **SUBMIT**. A green **Success** message displays, and the new downlink appears in the list.

NOTE: You might need to click **REFRESH** to see the LAG.

< Fabrics / Fabric: PowerStore Fabric

Switches **LAG** Topology Fabric Upgrade Downlinks Uplinks Health Settings

Success: Lag: 500T-nodeA-LAG created successfully.

REFRESH

+ ADD

EDIT

DELETE

	Name	Info	Lag Type/ Mode	Admin State
☐ >>	500T-nodeA-LAG	Interfaces 2	McLag LACP	up

LAGs per page 10 1 - 1 of 1 LAGs

Figure 28. Fabric LAG tab – LAG created

Repeat the above steps for each LAG on the cluster network from the fabric leaf switches to the PowerStore appliances. There should be two LAGs per PowerStore appliance for the cluster network.

For this example, there are two PowerStore appliances (named 500T and 1000T). When complete, the **LAG** page appears as shown in the figure below.

Fabrics / Fabric: PowerStore Fabric

Switches
LAG
Topology
Fabric Upgrade
Downlinks
Uplinks
Health
Settings

Success: Lag: 1000T-nodeB-LAG created successfully.

REFRESH
+ ADD
EDIT
X DELETE

	Name	Info	Lag Type/ Mode	Admin State
☐	500T-nodeA-LAG	Interfaces 2	McLag LACP	up
☐	500T-nodeB-LAG	Interfaces 2	McLag LACP	up
☐	1000T-nodeA-LAG	Interfaces 2	McLag LACP	up
☐	1000T-nodeB-LAG	Interfaces 2	McLag LACP	up

LAGs per page 10
1 - 4 of 4 LAGs

Figure 29. LAGs to PowerStore appliances created

After creating the LAGs, they are placed in the PowerStore cluster network as follows:

1. Click the **Downlinks** tab.
2. Click **+ADD**.
3. In the **Create Downlink** dialog box, do the following:
 - a. Enter a **Name** for the downlink.
 - b. Next to **Tenant**, select the nondefault tenant that you created in [Create a nondefault tenant](#).
 - c. Set the **Interface Type** to **LAG** and select the **LAG** that you created for the PowerStore appliance node.
 - d. Next to **Untagged Network**, select the cluster VLAN. PowerStore appliance cluster interfaces are untagged by default.

Create Downlink

Name 500T-nodeA-cluster
Tenant PS-tenant x

Port Configuration

Interface Type
☒ LAG
☐ Physical

LAG
500T-nodeA-LAG x

Network Configuration
+ ADD NETWORK

Untagged Network
cluster-1 x

Tagged Network
Select Tagged Network

CANCEL
SUBMIT

Figure 30. Downlink dialog box for PowerStore cluster link

- e. Click **SUBMIT**. A green **Success** message displays, and the new downlink appears in the list.

< Fabrics / Fabric: PowerStore Fabric

Switches LAG Topology Fabric Upgrade **Downlinks** Uplinks Health Settings

Success: Downlink: 500T-nodeA-cluster created successfully.

REFRESH + ADD EDIT DELETE

	Name	Info	Tenant Name	Status
☐ >>	500T-nodeA-cluster	Networks 1 Interfaces 1	PS-tenant	InProgress
☐ >>	500T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	500T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	1000T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	1000T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC01-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC01-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC02-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐ >>	SC02-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	Success

Downlinks per page 10 1 - 9 of 9 Downlinks

Figure 31. Fabric downlinks tab – PowerStore cluster downlink added

When SFM has configured the downlink on the switches, the **Status** changes from **In Progress**, as shown, to **Success**. The status might take some time to change. You can click **REFRESH** as needed to update the status. You do not have to wait for the status to change before configuring the next downlink.

Repeat the above steps for each link from the fabric leaf switches to PowerStore interfaces that are on the cluster network.

For this example, there are two PowerStore appliances (named 500T and 1000T). When the cluster network downlinks are complete, the **Downlinks** tab appears as shown in the figure below. The cluster downlinks are outlined in red.

< Fabrics / Fabric: PowerStore Fabric

Switches LAG Topology Fabric Upgrade **Downlinks** Uplinks Health Settings

Success: Downlink: 1000T-nodeB-cluster created successfully.

REFRESH + ADD EDIT X DELETE

	Name	Info	Tenant Name	Status
○ >>	500T-nodeA-cluster	Networks 1 Interfaces 1	PS-tenant	InProgress
○ >>	500T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
○ >>	500T-nodeB-cluster	Networks 1 Interfaces 1	PS-tenant	Success
○ >>	500T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
○ >>	1000T-nodeA-cluster	Networks 1 Interfaces 1	PS-tenant	InProgress
○ >>	1000T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
○ >>	1000T-nodeB-cluster	Networks 1 Interfaces 1	PS-tenant	InProgress
○ >>	1000T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
○ >>	SC01-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
○ >>	SC01-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	Success

Downlinks per page 10 1 - 10 of 12 Downlinks 1 / 2

Figure 32. PowerStore cluster downlinks configured

Configure PowerStore appliances

If not already done, make the PowerStore appliance connections to the switches as shown in [Production topology connection details](#) and [OOB management network connection details](#).

Topics:

- [Initial configuration](#)
- [NVMe/TCP network configuration](#)
- [Cluster network](#)

Initial configuration

Connect a workstation to the PowerStore appliance using the OOB management switch and run the PowerStore Initial Configuration Wizard (ICW) as shown in the [Dell PowerStore T and Q Models Networking Guide for Initial Deployment](#).

NVMe/TCP network configuration

Use PowerStore Manager to configure the NVMe/TCP network on the PowerStore appliance as shown in the [Dell PowerStore PowerStore T and Q Networking Guide for Storage Services](#).

Specify the same VLAN ID in PowerStore Manager that is used in the SFM fabric (VLAN 300 in this example). Set the MTU size for the NVMe/TCP network to 9000 bytes in PowerStore Manager for best performance.

 **NOTE:** The MTU size is automatically set to 9216 bytes on all leaf and spine interfaces by SFM.

Cluster network

If you will be using a PowerStore cluster, the cluster network is configured in SFM as shown in this guide. No further configuration is required in PowerStore Manager as the PowerStore cluster network interfaces are preconfigured.

Verify the infrastructure

SFM can be used to verify that the fabric and node connections are configured and working properly. This chapter shows how to validate the fabric for this deployment.

1. In the SFM UI, click **Fabric** in the left pane. The **Managed Fabrics** page opens in the right pane.

Managed Fabrics

[REFRESH](#)
[EDIT](#)
[DELETE](#)

	Fabric Name	No. of Leaves	No. of Spines	No. of SuperSpines	No. of BorderLeaves	Fabric Type	Site	Action
☐	PowerStore Fabric	4	2	0	0	VxLan	1	MANAGE

Fabrics per page 10 1 - 1 of 1 Fabrics

Figure 33. Managed Fabrics page

2. Next to the PowerStore fabric name, click **MANAGE**. The **Fabric: *Fabric Name*** page opens, and the **Switches** tab displays.

< Fabrics / Fabric: PowerScale Fabric

[Switches](#)
[LAG](#)
[Topology](#)
[Fabric Upgrade](#)
[Downlinks](#)
[Uplinks](#)
[Health](#)
[Settings](#)

[REFRESH](#)
[DELETE](#)
[CLAIM OWNERSHIP](#)
[RUN CLI COMMANDS](#)

☐	Name	Role	IP Address	Status	Mac Address
☐ >>	leaf1a	Leaf	100.67.112.13	Online	54:BF:64:E6:DD:C1
☐ >>	leaf1b	Leaf	100.67.112.14	Online	3C:2C:30:10:35:00
☐ >>	leaf2a	Leaf	100.67.112.15	Online	3C:2C:30:89:45:80
☐ >>	leaf2b	Leaf	100.67.112.16	Online	3C:2C:30:22:79:80
☐ >>	spine1	Spine	100.67.112.17	Online	8C:04:BA:B7:C5:C0
☐ >>	spine2	Spine	100.67.112.18	Online	8C:04:BA:B7:9A:40

Nodes per page 10 1 - 6 of 6 Nodes

Figure 34. Fabrics page - Switches tab

3. Ensure that all leaf and spine switches in the fabric are listed and are **Online**.
4. To view more information, click the >> next to a switch name. In the new pane that opens to the right, there are tabs that are named **Switch Details**, **Interfaces**, and **Health**, outlined in red in the figure below.

< Fabrics / Fabric: PowerStore Fabric

Switches LAG Topology Fabric Upgrade Downlinks Uplinks Health Settings

REFRESH X DELETE CLAIM OWNERSHIP RUN CLI COMMANDS

Name	Switch Details	Interfaces	Health																									
leaf1a	REFRESH EDIT	<table border="1"> <thead> <tr> <th>Name</th> <th>Admin Status</th> <th>Operational Status</th> <th>MTU</th> <th>Speed (Mbps)</th> </tr> </thead> <tbody> <tr> <td>Eth1/1</td> <td>up</td> <td>up</td> <td>9216</td> <td>25000</td> </tr> <tr> <td>Eth1/2</td> <td>up</td> <td>up</td> <td>9216</td> <td>25000</td> </tr> <tr> <td>Eth1/3</td> <td>up</td> <td>up</td> <td>9216</td> <td>25000</td> </tr> <tr> <td>Eth1/4</td> <td>up</td> <td>down</td> <td>9216</td> <td>25000</td> </tr> </tbody> </table>	Name	Admin Status	Operational Status	MTU	Speed (Mbps)	Eth1/1	up	up	9216	25000	Eth1/2	up	up	9216	25000	Eth1/3	up	up	9216	25000	Eth1/4	up	down	9216	25000	
Name	Admin Status	Operational Status	MTU	Speed (Mbps)																								
Eth1/1	up	up	9216	25000																								
Eth1/2	up	up	9216	25000																								
Eth1/3	up	up	9216	25000																								
Eth1/4	up	down	9216	25000																								
leaf1b																												
leaf2a																												
leaf2b																												
spine1																												
spine2																												

Figure 35. Switch Interfaces for leaf1a

- The **Switch Details** tab provides information such as the model, service tag, and SONiC software version.
 - The **Interfaces** tab, selected above, shows port **Operational Status** and other settings. If the switches and PowerStore appliances are configured properly, the port **Operational Status** should be **up** for the connected ports. Individual ports can be selected and edited to change settings such as admin state and MTU. Other settings might also be available depending on the PowerSwitch model.
 - The **Health** tab provides software and hardware health scores for the switch.
5. At the top of the **Fabrics** page, click the **LAG** tab. All LAGs in the topology are listed. Single PowerStore appliance deployments will not have LAGs for PowerStore. If you are using a PowerStore cluster, verify that each PowerStore appliance LAG has two **Interfaces**, is configured for **MC-LAG** and **LACP**, and the **Admin State** is **up** as shown in the figure below.

< Fabrics / Fabric: PowerStore Fabric

Switches LAG Topology Fabric Upgrade Downlinks Uplinks Health Settings

REFRESH + ADD EDIT X DELETE

Name	Info	Lag Type/ Mode	Admin State
500T-nodeA-LAG	Interfaces 2	McLag LACP	up
500T-nodeB-LAG	Interfaces 2	McLag LACP	up
1000T-nodeA-LAG	Interfaces 2	McLag LACP	up
1000T-nodeB-LAG	Interfaces 2	McLag LACP	up

LAGs per page 10 1 - 4 of 4 LAGs

Figure 36. LAG tab

- For more details, click the **>>** next to a PowerStore LAG name. The **LAG Details** pane opens to the right.
- Click the **>** next to **Interfaces**. The **Switch Name**, **Interface**, and **Port Channel ID** for the LAG is shown.
- Make sure that the **Operational State** is **up** on both leafs, outlined in red in the figure below.

< [Fabrics](#) / Fabric: PowerStore Fabric

Switches **LAG** Topology Fabric Upgrade Downlinks Uplinks Health Settings

REFRESH + ADD EDIT DELETE

	Name
☐ <<	500T-nodeA-LAG
☐ >>	500T-nodeB-LAG
☐ >>	1000T-nodeA-LAG
☐ >>	1000T-nodeB-LAG

1-4 / 4 < 1 >

LAG Details

LAG ID: 7a9bb325-673a-4f6a-9095-098eb42ea126

Interfaces 2

Switch Name	Interface	PortChannel Id	Operational State
leaf1a	Eth1/11	255	up
leaf1b	Eth1/11	255	up

Lag Type / Mode: McLag LACP

Rack ID: cddadd80-6490-4ec9-af2a-448d4a0ba54e

Fabric ID: edf36843-bb41-419f-8fa5-a5772a478975

Figure 37. LAG details

- Click the **Topology** tab at the top of the **Fabrics** page. The fabric topology is shown with leaf and spine connections. A missing connection indicates that a link is down or is not physically connected.

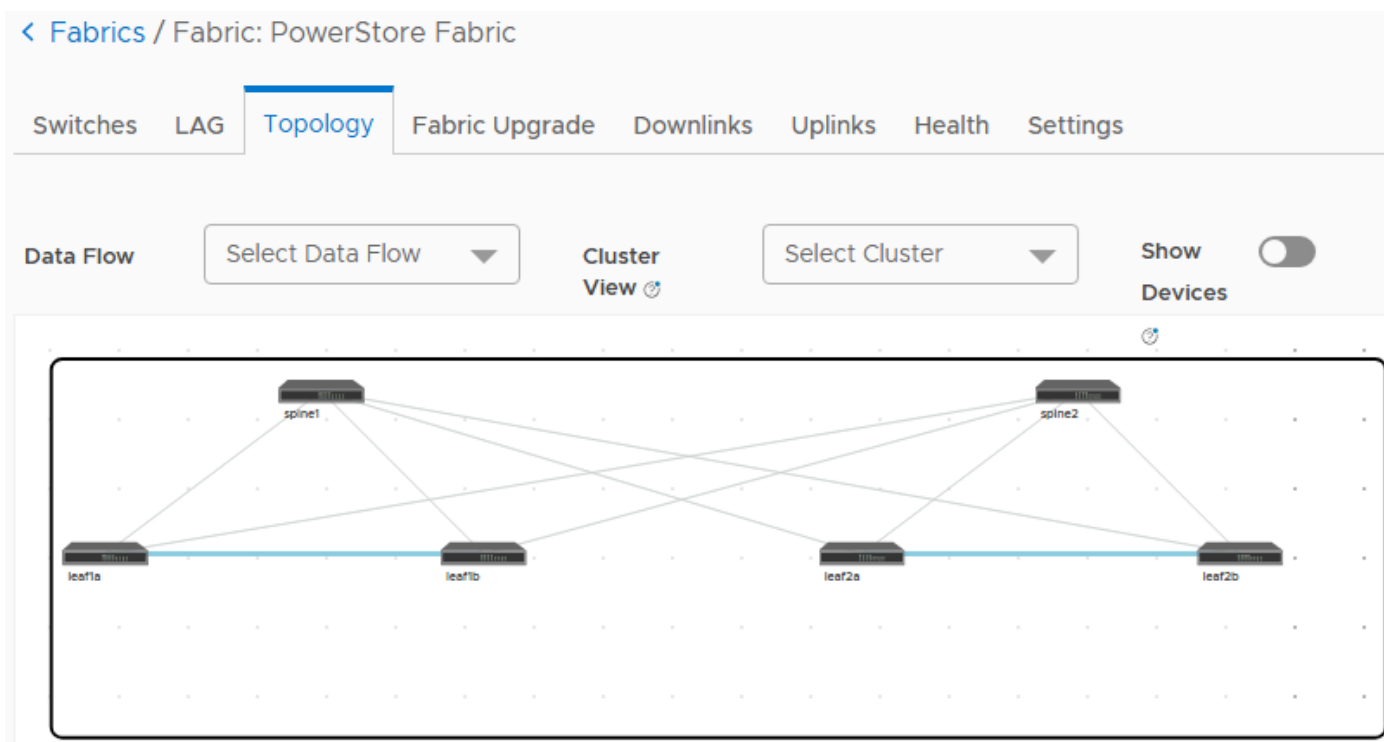


Figure 38. Topology view with PowerStore appliances

- Click the **Downlinks** tab at the top of the **Fabrics** page. All downlinks in the topology are listed.

REFRESH + ADD EDIT DELETE

		Name	Info	Tenant Name	Status
☐	>>	500T-nodeA-cluster	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	500T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	500T-nodeB-cluster	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	500T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	1000T-nodeA-cluster	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	1000T-nodeA-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	1000T-nodeB-cluster	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	1000T-nodeB-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	SC01-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	SC01-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	SC02-p1-NVMe	Networks 1 Interfaces 1	PS-tenant	Success
☐	>>	SC02-p2-NVMe	Networks 1 Interfaces 1	PS-tenant	Success

Downlinks per page 20 1 - 12 of 12 Downlinks

Figure 39. Downlinks tab

11. Click the >> icon next to a downlink name to view additional details. The **Downlink Details** pane displays on the right.
 - a. Click the > next to **Tagged Networks** to view the VLANs that the port is tagged in. PowerStore appliance NVMe downlinks must be tagged in the NVMe VLAN as shown in the figure below.
 - b. If it is untagged in a VLAN, verify that the VLAN is listed under **Untagged Network**. PowerStore appliance cluster downlinks must be untagged in the cluster VLAN.
 - c. Click the > next to **Interfaces**. If the downlink is not in a LAG, such as an NVMe downlink, the switch interface is listed. If the downlink is in a LAG, such as a cluster downlink, the LAG name is listed.

< [Fabrics](#) / Fabric: PowerStore Fabric

Switches LAG Topology Fabric Upgrade **Downlinks** Uplinks Health Settings

REFRESH + ADD EDIT DELETE

	Name
☐	500T-nodeA-cluster
☒	500T-nodeA-NVMe
☐	500T-nodeB-cluster
☐	500T-nodeB-NVMe
☐	1000T-nodeA-cluster
☐	1000T-nodeA-NVMe
☐	1000T-nodeB-cluster
☐	1000T-nodeB-NVMe
☐	SC01-p1-NVMe
☐	SC01-p2-NVMe

1-10 / 12 < 1 >

Downlink Details

Downlink Id	ac0bb5dc-e20c-4e94-a1a1-31e1c66b3f00				
Tenant Id	43ff7692-1aba-496f-b73d-e574084c0cd2				
Tenant Name	PS-tenant				
Tagged Networks 1 <table border="1"> <thead> <tr> <th>Name</th> <th>VLAN</th> </tr> </thead> <tbody> <tr> <td>NVMe-300</td> <td>300</td> </tr> </tbody> </table>		Name	VLAN	NVMe-300	300
Name	VLAN				
NVMe-300	300				
Untagged Network					
Interfaces 1 <table border="1"> <thead> <tr> <th>Switch Name</th> <th>Interface Name</th> </tr> </thead> <tbody> <tr> <td>leaf1a</td> <td>Eth1/10</td> </tr> </tbody> </table>		Switch Name	Interface Name	leaf1a	Eth1/10
Switch Name	Interface Name				
leaf1a	Eth1/10				
Interfaces per page 10 1 - 1 of 1 Interfaces					
Status	Success				

Figure 40. NVMe downlink details

12. Click the **Health** tab to display the fabric health

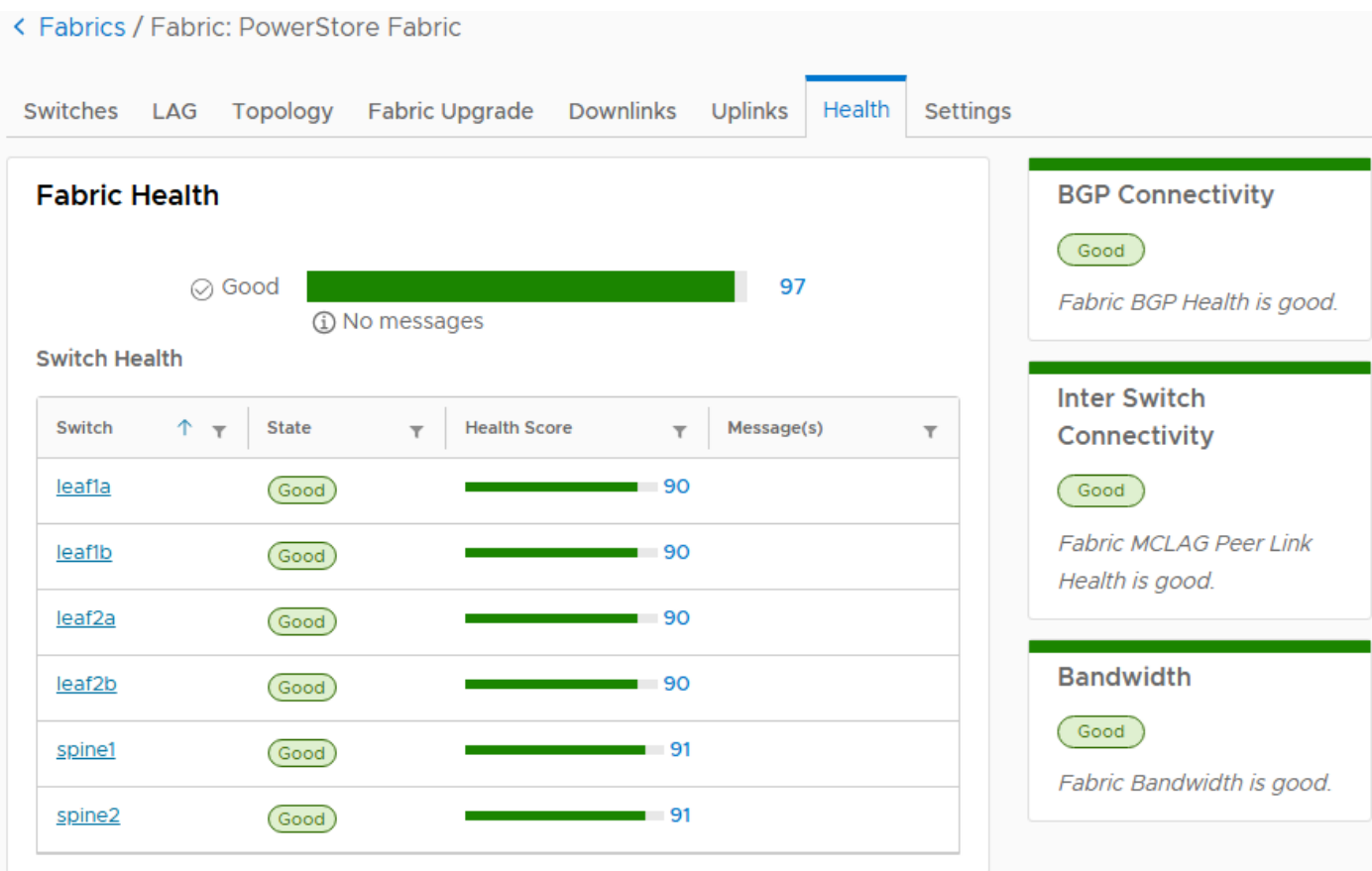


Figure 41. Fabric Health tab

- Ensure that the **State** is **Good**, and that the **Health Score** is green for each switch in the fabric. If any switches have a yellow or red **Health Score**, see the **Messages** column for the cause.

Resources

Topics:

- [Dell Networking resources](#)
- [PowerStore resources](#)
- [Technical support](#)

Dell Networking resources

[Dell Technologies Networking Solutions Info Hub](#) – This site provides guides to assist with planning and deploying various solutions with Dell Networking.

[Dell Technologies Enterprise SONiC Distribution](#) (account required) – This site includes SONiC user guides and software.

[SmartFabric Manager documentation](#) (account required)

[SmartFabric Manager software](#) (account required)

[Dell PowerSwitch S3248T-ON Documentation](#)

[Dell PowerSwitch S5212F-ON Documentation](#)

[Dell PowerSwitch S5248F-ON Documentation](#)

[Dell PowerSwitch S5232F-ON Documentation](#)

[Dell Networking Transceivers and Cables](#)

PowerStore resources

[Dell PowerStore T and Q Models Networking Guide for Initial Deployment](#)

[Dell PowerStore T and Q Networking Guide for Storage Services](#)

[Dell Networking Switch Simple Support Matrix](#) (account required)

[Dell PowerStore: Introduction to the Platform](#)

[PowerStore Product Documentation](#)

Technical support

For technical support, go to [the Dell Support site](#).

Validated components

The following tables include the hardware, software, and firmware that was used to validate the example in this guide. Other supported hardware, software, and firmware can be used.

Topics:

- [Dell PowerSwitch systems](#)
- [SONiC SmartFabric Manager](#)
- [PowerStore appliance software](#)
- [PowerEdge servers](#)

Dell PowerSwitch systems

The switches used to validate the deployment in this guide are listed in the table below.

Table 4. Dell PowerSwitch models and operating systems

Qty	Item	Role	SONiC version
4	Dell PowerSwitch S5248F-ON	Leaf switch	4.4.1 Enterprise Standard
2	Dell PowerSwitch S5232F-ON	Spine switch	4.4.1 Enterprise Standard
1	Dell PowerSwitch S3248T-ON	OOB management switch	4.4.1 Lite

SONiC SmartFabric Manager

A Dell PowerEdge server running VMware ESXi was used as the host for the SONiC SmartFabric Manager virtual machine.

Table 5. SmartFabric Manager software

Item	Version
SONiC SmartFabric Manager	1.2.0.0213
VMware ESXi host for SFM	8.0.3 build 24280767 Dell customized A02

PowerStore appliance software

One PowerStore 500T appliance and one PowerStore 1000T appliance were used to provide NVMe/TCP storage volumes.

Table 6. PowerStore appliance configuration

Qty per appliance	Item	Version
1	PowerStore appliance software	4.0.0.2
2	4-port card, 4x 25 GbE SFP28	-
2	IO module, 4x 25 GbE SFP28	-

PowerEdge servers

Two Dell PowerEdge R640 servers running VMware ESXi were used as NVMe/TCP storage clients.

Table 7. PowerEdge server configuration

Qty per server	Item	Version
1	Operating system	VMware ESXi 8.0.3 build 24280767 Dell customized A02
1	Broadcom 25 GbE SFP28 dual-port adapter	Firmware 22.92.07.50

BGP EVPN with VXLAN overview

BGP EVPN with VXLAN stretches virtual (overlay) Layer 2 networks across a physical (underlay) Layer 3 leaf-spine fabric. This architecture allows for the scalability of Layer 3 networks with the mobility benefits of a Layer 2 network. For example, a VM or physical host can be moved from one rack to another without the need to change its IP address and gateway information. All paths in the fabric are active and no connections are blocked by the spanning tree protocol.

EVPN uses BGP to exchange endpoint MAC and IP address information between VXLAN tunnel endpoints (VTEPs). In this topology, each pair of leaf switches in a rack is one logical VTEP. Packets destined to a remote device are sent to the local VTEP as the first hop. The packet is encapsulated with a VXLAN header that includes the Virtual Network Identifier (VNI) and the IP address of the destination host. See the figure below.

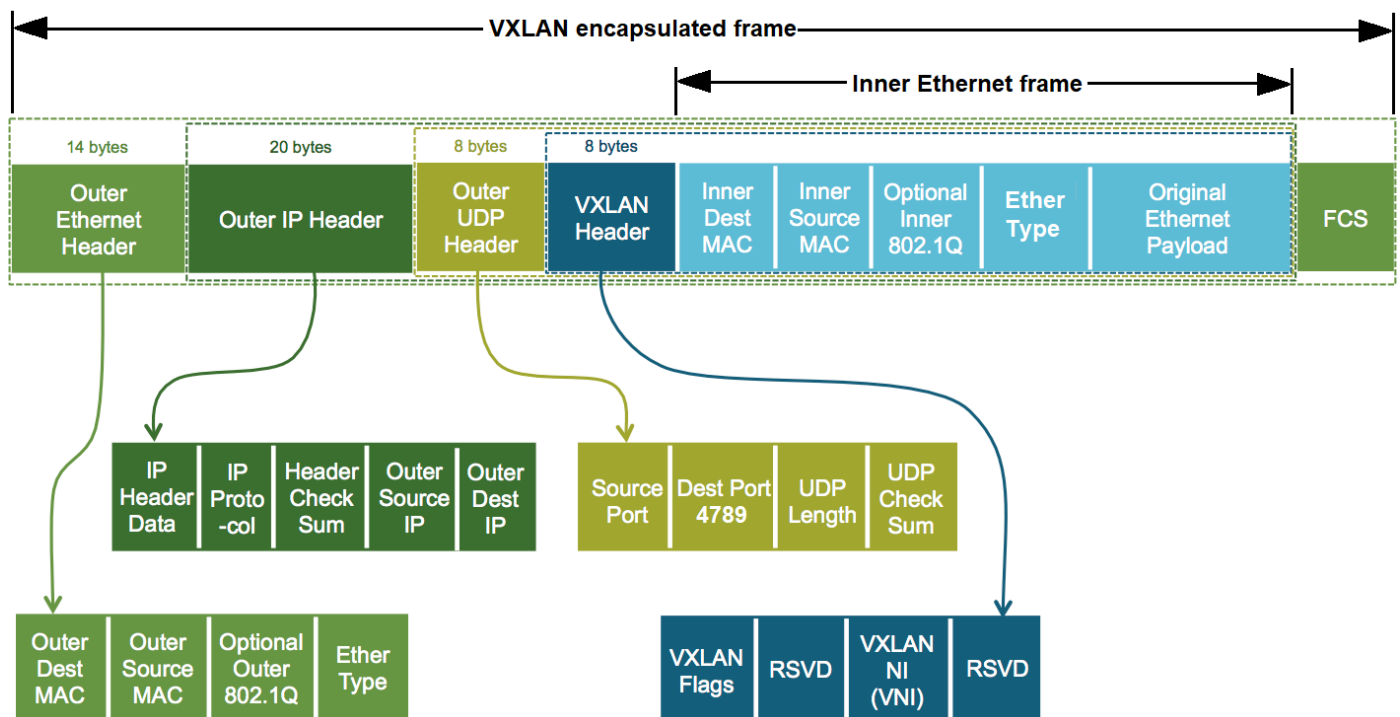


Figure 42. VXLAN encapsulated frame

The destination VTEP extracts the original packet and routes it to the destination. VXLAN encapsulation adds approximately 50 bytes of overhead to each Ethernet frame. As a result, all switches in the underlay (physical) network must be configured to support an MTU of at least 1600 bytes on all participating interfaces.

NOTE: In this deployment, switch interfaces are automatically set to their maximum supported MTU size of 9216 bytes by SFM for best performance.

Anycast gateways with the same IP address are configured on each leaf pair. An anycast gateway IP address is set as the default gateway for hosts and VMs on that virtual network. Hosts and VMs on the same virtual network use the same gateway address while behind different leaf pairs. This configuration replaces the Virtual Router Redundancy Protocol (VRRP) and enables hosts and VMs to migrate from one leaf pair to another without changing the network configuration.